Outdoor mean radiant temperature estimation: is the black-globe thermometer method a feasible course of action?

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***Abstract*— Urban planning has become one of the crucial action in order to address and mitigate the effect and the consequences of the global warming and it also entails different sectors of the urban context.**

The survey and control of the outdoor climatic conditions are among the involved sectors and mean radiant temperature is amid the most analyzed parameters.

The aim of the paper is to investigate the effective suitability of one of the most used method in the assessment of the mean radiant temperature, namely the globe-thermometer method, in outdoor urban environment, singling out possible shortcomings and actual efficacy.

Specifically, data obtained by measurement campaigns performed by means of a black-globe thermometer were compared with the ones resulted by the application of the six-directional technique based on the measure of the radiative fluxes (longwave and shortwave radiation) coming from six perpendicular space directions.

Keywords—Mean radiant temperature;outdoor environment long wave radiation; sola radiation; globe termometer;

I. INTRODUCTION

Urban settlements are deemed to be among the human activities which exert the most deeply spoiling impacts on the environment [1].

Being responsible of a large share of the European energy consumption and one of the main sources of the environmental pollution, they are the pivotal objective of the European strategies concerning sustainability [2] [3].

In this direction the "Covenant of Mayors for Climate & Energy" [4] acts: it commits local and regional authorities, which voluntary adhere, to implementing actions and strategies aimed at restraining energy consumption and environmental pollution on their territory, thus contributing to pursuing the EU targets in terms of emission reduction and control of the climate change processes.

In this framework, urban planning has become one of the crucial action in order to address and mitigate the effect and the consequences of the global warming and it also entails different sectors of the urban context [5], [6].

The survey and control of the outdoor climatic conditions are among the involved sectors [7], [8]. As a matter of fact, not only do they affect the energy consumption of buildings and the chance of renewable source exploitation [9], but they also determine the quality of outdoor urban space and might influence various human behaviors [10] and related activities as pedestrian traffic and modal choices of the transportation systems, whose rationalization is one of the crucial objective of the action plans adopted by local authorities in view of tackling mitigation and adaptation to climate change. [11]

The structure and configuration of the urban settlements have acknowledged to be able to influence the outdoor microclimate: the intensity of the heat island phenomenon (UHI), for instance, is affected by the morphology of the urban areas and therefore its mitigation should be part of present and future design strategies.

In this context, thermal comfort and stress assessment of the urban sites [12] can be exploited as additional information to standard UHI analyses, yielding crucial information about the thermal quality of the outdoor environment [13] .

From this point of view, one of the main aspect that needs to be accurately assessed regards the radiant field with which the human subject interacts and which depends on the temperature, shape and position of the surrounding surface (longwave radiation sources), and on the presence of the solar irradiance (shortwave radiation) [14], [15].

Combining the thermal effect of short and long wave radiation fluxes from the surroundings, the mean radiant temperature, MRT, is one of the main factor used to quantify the effect of the radiant field on human thermal response.

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In outdoor environment and especially in urban areas, its assessment is complicated by the presence of heterogeneous radiant conditions, solar radiation shining directly on the subject but also on the surrounding surfaces and influencing their temperatures, shading effects and morphology of terrain and landscape.

However, several methods are available in literature, but, more than often, given the complexity which characterizes urban areas, they would require a large effort in terms of time and devices involved.

The purpose of this paper is to investigate on the effective suitability of one of the most used measurement method, namely the globe-thermometer method, in outdoor urban environment, singling out possible shortcomings and actual efficacy.

Specifically, data obtained by a measurement campaign performed by means of a black-globe thermometer were compared with the ones resulted by the application of the six-directional technique based on the measure of the radiative fluxes (longwave and shortwave radiation) coming from six perpendicular space directions, with a view to evaluating the actual feasibility of the method.

II. Methods for Mean Radiant Temperature Assessment

Mean Radiant Temperature ($\bar{T}_r$) is the uniform temperature of an imaginary enclosure in which radiant heat transfer from the human body is equal to the radiant heat transfer in the actual non-uniform enclosure [16].

The estimation of $\bar{T}_r$ can be performed by means of different methods [16], but in urban settlements, its assessment may result in a very articulated procedure owing to the complexity of the enveloping environment, space and time variability of the parameter, number of factors and variables to be considered [17]–[19].

In this frame, the proposed analysis focuses on two specific methods which are based on measurements of parameters characterizing the radiant field: the ISO 7726 globe-thermometer method [16] and the six-dimensional technique, introduced by Höppe [20]. The latter is considered one of the most accurate measurement to obtain outdoor $\bar{T}_r$ [21], but it is quite expensive and rather immobile, because of the needed equipment.

On the contrary, considering the time and space variability of the involved parameters, the use of a more streamlined procedure would be desirable in order to make the process of MRT assessment more feasible, especially on large outdoor areas.

The proposed analysis aims to investigate the reliability of the ISO 7726 globe-thermometer method in outdoor urban environments. To fulfill the task comparison with the six-dimensional technique were performed.

In addition, an analytical model [22], [23] was considered, with a view to analyzing its consistency, by comparing its outcome to the experimental results.

Specifically, the model is an adaptation of the methods formerly elaborated by the authors [22], [23], in order to make them suitable for outdoor environments.

All the cited methods are reported in the following sections.

A. The globe-thermometer method

The black-globe thermometer consists of a black globe in the centre of which is placed a temperature sensor such as the bulb of a mercury thermometer, a thermocouple or a resistance probe [16].

The globe tends towards a thermal balance under the effect of the exchanges due to the radiation coming from the different heat sources of the enclosure and under the effect of the exchanges by convection.

The globe temperature represents the weighted average of radiant and ambient temperatures. If the globe temperature, air temperature and air velocity are known then the $\bar{T}_r$ (K) can be calculated according to Equation [16]:

$$\bar{T}_r = \sqrt[4]{T_g^4 + \frac{h_c}{\varepsilon_g \sigma}\left(T_g - T_a\right)} \tag{1}$$

where: T_g is the globe temperature (K); h_c is the convective coefficient $\left(\frac{W}{m^2 K}\right)$; ε_g is the emissivity of the globe; σ is the Stefan-Boltzman constant $\left(\frac{W}{m^2 K^4}\right)$; T_a is the air temperature (K).

Furthermore, in natural convection:

$$h_c = 1.4\left(\frac{T_g - T_a}{D}\right)^{\frac{1}{4}} \tag{2}$$

whereas in forced convection:

$$h_c = 6.3\,\frac{v_a^{0.6}}{D^{0.4}} \tag{3}$$

where: D is the globe diameter (m); v_a is the air velocity (m/s).

The maximum value between the ones yielded by equations (2) and (3) is to be substituted in (1) [16].

B. The six-dimensional technique

The method simplifies the surrounding radiant field considering the radiative heat fluxes coming from six directions [21] four of which point to the four cardinal points of the compass, whereas the remaining two lie on the vertical plane and point to the sky and to the ground, respectively.

Under this hypothesis the mean radiative flux density absorbed by the human body, Q_{abs}, may be assessed as [20], [21]:

$$Q_{abs} = \alpha_s \sum_{1=1}^{6} F_i\, Q_{SW_i} + \varepsilon_p \sum_{i=1}^{6} F_i\, Q_{LW_i} \tag{4}$$

where: α_s is the absorption coefficient for short-wave radiation; F_i directional weighting factor related to direction i; Q_{SW_i} is the short-wave radiaction flux from direction i (W m^{-2}); ε_p is the

emissivity of the human body; $Q_{LW\,i}$ is the long-wave radiation flux from direction i (W m^{-2}).

The absorption coefficient for short-wave radiation, α_s, depends on the radiative source properties; when solar radiation is involved, its value is set to 0.7 [24].

On the other hand, the emissivity of the human body, ε_p, is commonly set to 0.97 [25].

F_i depends on the position and orientation [25]; its calculation is complicated for complex urban forms and simplifications are thus necessary. For a rotationally symmetric standing or walking person, F_i is set to 0.22 for radiation fluxes from the four cardinal points (east, west, north and south) and 0.06 for radiation fluxes from above and below. For a sphere, F_i is 0.167 for all six directions [17], [21].

Once Q_{abs} is known, the mean radiant temperature, $\bar{T}_r$ (K), may me calculated by means of the Stefan-Boltzmann law:

$$\bar{T}_r = \sqrt[4]{\frac{Q_{abs}}{\varepsilon_p\,\sigma}} \tag{5}$$

C. Analytical model

The model here proposed is an adaptation of the methods formerly elaborated by the authors [22], [23], in order to make them suitable for outdoor environments.

With a view to achieving this goal, it was assumed that the diffused component of solar radiation and the one reflected from the ground are distributed on the highest and lowest half of the exposed portion of the body, respectively. In this case, the mean radiant temperature can be calculated by means of the equation:

$$\bar{T}_r = \sqrt[4]{\frac{Q_{LW}}{\sigma} + \frac{\alpha_{SW}}{\varepsilon_p \sigma}\left\{0.5\left[I_{dh} + \rho_{grd}(I_{dh} + I_{bh})\right] + f_p I_b\right\}} \tag{6}$$

where: Q_{LW} is the global long-wave radiative flux coming from the enveloping environment (W m^{-2}); I_{dh} is the diffuse component of the solar radiation (short-wave) shining on the horizontal plane (W m^{-2}); I_{bh} is the drect component of the solar radiation (short-wave) shining on the horizontal plane (W m^{-2}); ρ_{grd} is the reflectance of the ground; I_b is the direct component of solar radiation on a plane orthogonal to the sunbeam (W m^{-2}); f_p is the projected area factor of the human body, calculated with reference to the direction of the sunbeam.

As regards the calculation of f_p, the models reported in [26], [27] can be exploited.

III. METHODOLOGY

The proposed analysis aims to compare the results yielded by the three methods of obtaining mean radiant temperature which are reported in the former sections.

To fulfill the task, a measurement campaign was performed. It regarded the short-wave and long-wave radiation fluxes coming from the six space directions, all the needed climatic parameters (air temperature and air velocity) and the globe temperature. Experimental apparatus features and measurement configurations are described in the following paragraphs.

A. Experimental apparatus

1) Radiant fluxes measurement

The instrument setup used for the radiant flux measurement is shown in Fig. 1.

Fig. 1 Instrument setup used for the radiant flux measurements.

It consists of three probes installed on a pole. Each probe is a CNR4™ net radiometer (Kipp and Zonen), which, in turn, consists of a pyranometer pair, disposed on the opposite surfaces of the probe's plane, and a pyrgeometer pair in a similar configuration. The pyranometer pair measures the short-wave radiation, whereas the pyrgeometer pair measures long-wave radiation.

The measured data are collected through a set of two loggers (QML102™, VAISALA) which are connected to a laptop, located in a nearby container, through a Ethernet port.

2) Climatic parameter measurement

Air temperature and velocity were measured by means of the Vaisala MAWS201 portable weather station equipped with the WM302 wind sensor and the HMP155 humidity and temperature probe (Fig. 2).

a) WM302 wind sensor b) HMP155 humidity and temperature probe

Fig. 2 wind and temperature sensors

Measured data were collected by means of a QML102™ logger (Vaisala) and periodically downloaded through an Ethernet port.

3) Globe temperature measurement

The experimental apparatus (Fig. 3) is a monitoring system consisting of a programmable data-logger LSI Lastem E-log to which a EST131™ black globe -thermometer is connected.

The diameter D of the globe is 0.15m long, whereas it emissivity is equal to 0.95.

Fig. 3 Instrument setup used for the globe temperature measurements.

B. Measurement configuration and data treatment

The measurement campaign took place at the Mediterranean University campus (38°07'14.3" North latitude, 15°39'57.7" Est longitude). The measure point (P, Fig. 4) is located inside the University courtyard which is equipped with all the needed networks.

The measures were performed during various periods from 01/03/2018 to 16/03/2018.

During those periods, both the 10 minute and 1 minute average values of the parameters were detected and gathered.

As far as the solar radiation is concerned, it is worth noting that, for the application of the analytical model based on equation (6), both the diffuse and direct component are needed.

On the contrary, the used CNR4™ net radiometer measures the global solar radiation on the horizontal plane.

Therefore, the two components need to be assessed and with a view to reaching this aim, the method reported in [28] was used. Specifically:

$$\frac{I_{dh}}{I_{bh}+I_{dh}} = \begin{cases} 1.020 - 0.248\,k_t & if \quad 0.0 \le k_t \le 0.3 \\ 1.450 - 1.670\,k_t & if \quad 0.3 < k_t < 0.78 \\ 0.147 & if \quad 0.78 \le k_t \end{cases} \quad (8)$$

where k_t is the ratio of global horizontal to extraterrestrial radiation and $(I_{bh} + I_{dh})$ is the measured global solar radiation on the horizontal plane.

In addition, the direct component of solar radiation on a surface orthogonal to the sunbeam, I_b, was calculated as:

$$I_b = \frac{I_{bh}}{sen\ \alpha} \quad (7)$$

where α is the solar altitude.

Fig. 4 Mediterranean University campus – Measurement point P.

As regards the term Q_{LW}, also needed for the solution of Eq. 6, it was calculated by means of the subsequent Eq. 8, exploiting the long-wave radiative fluxes, Q_{LW_i}, measured using the apparatus described in section III.A.

Specifically, Q_{LW} may be calculated as follows:

$$Q_{LW} = \sum_{i=1}^{6} F_i\, Q_{LW_i} \quad (8)$$

In short, in order to assess the global long-wave radiative flux coming from the enveloping environment (Q_{LW}), to be used in eq. (6), the six-dimensional technique was exploited.

Consequently, in this case, the comparison between the two methods (analytical model and six-dimensional technique) exclusively regarded the different approaches which they use to model the effect of the solar radiation.

IV. Results and Discussion

Results of the measurement campaign are reported in Fig. 4-8 for the period 14/03/2018-16/03/2018.

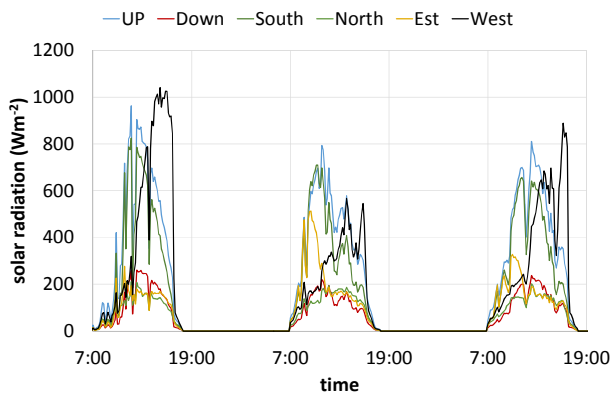

Fig. 5 Solar radiation measured within the period 14/03/2018-16/03/2018.

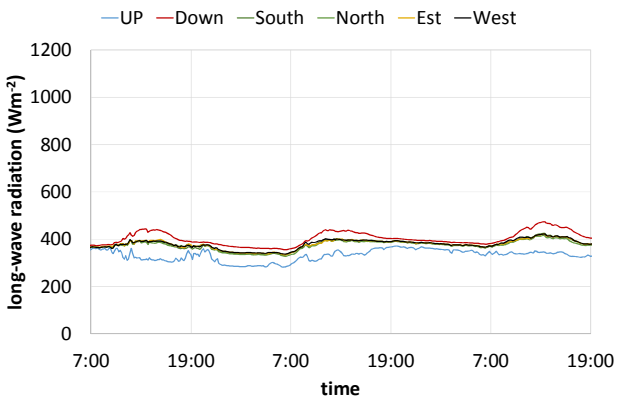

Fig. 6 Long-wave radiation measured within the period 14/03/2018-16/03/2018.

In this figures, all the reported data are averaged values over a time interval of 10 minutes. Therefore, each graph depicts the time trend of the 10 minute average of the correspondent parameter.

It may be noted that the chosen period is characterized by a rather high level of solar radiation, whose effect on mean radiant temperature may be, hence, properly analyzed.

On the other hand, wind speed remains lower than 3 m/s and air temperature ranges between 12°C and 22°C, so that, as a consequence, globe temperature can cycle between 5°C and 37°C.

This variability allows an in-depth analysis of the mean radiant temperature behavior.

Mean radiant temperature trend, for the same period, is depicted in Fig. 9, where the outcomes of all the three formerly described methods are reported.

It is worthy of note that the ISO 7726 method and six-dimensional technique yielded MRT values which have similar time trend, during both daylight and night periods.

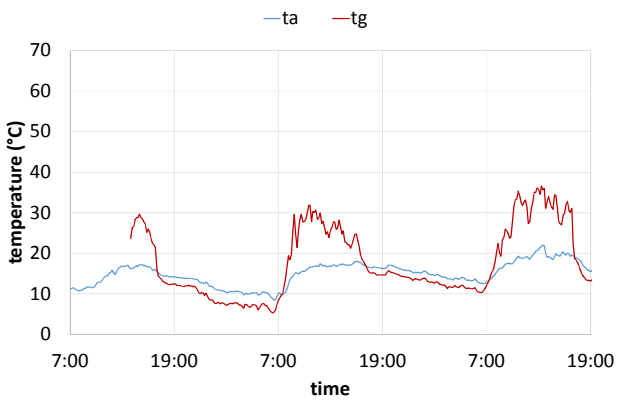

Fig. 7 Globe temperature (t_g) and air temperature (t_a) measured within the period 14/03/2018-16/03/2018.

Therefore, it may be inferred that the globe-thermometer measurement method is able to properly assess the time variability of the studied parameter, during the analyzed period

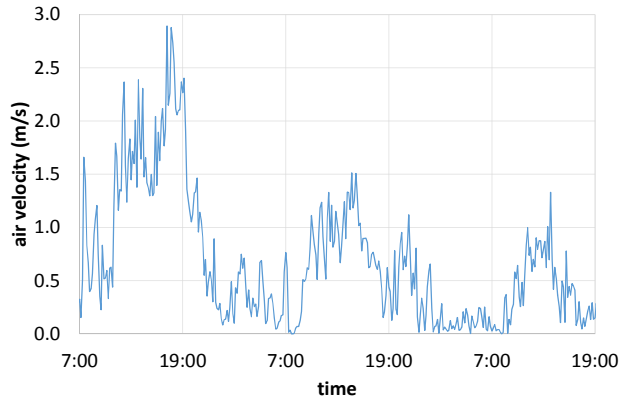

Fig. 8 Air velocity (v_a) measured within the period 14/03/2018-16/03/2018.

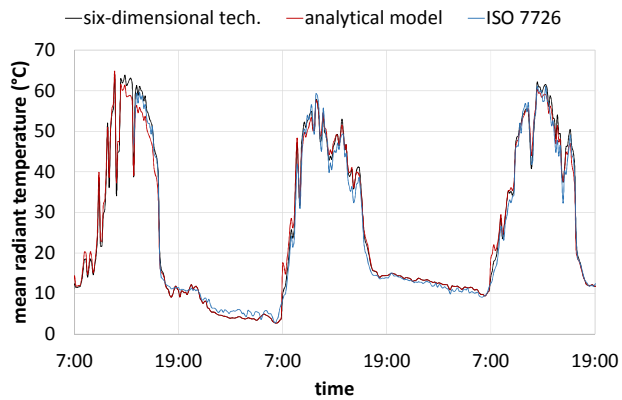

Fig. 9 Mean rediant temperature (t_r) evaluated within the period 14/03/2018-16/03/2018.

As regards the proposed analytical model, it must be stressed that the comparison, in this case, aims to verify its effectiveness in assessing only the share of MRT which is due to solar radiation (short-wave). From this perspective, the proposed analytical model also seems to yield results which are able to follow the time trend of the experimental outcomes.

Information regarding the reliability of both ISO 7726 method and the proposed analytical model may be also derived from Fig. 10 and Fig. 11, which regard the data assessed and measured within all the analyzed periods (01/03/2018-16/03/2018).

Those figures report the MRT values calculated by means of both ISO 7726 method and the proposed analytical model versus the results yielded by the application of the six-dimensional technique.

Specifically, Fig. 10 refers to 10 minute average values, whereas Fig. 11 concerns 1 minute average values.

As regards the analytical model results, only the data related to the daylight periods were reported, so that the comparison is limited to the portion of mean radiant temperature generated by solar radiation.

The analysis of the data shows that, especially when 10 minute average values are considered (Fig. 10), all the studied

approaches are consistent with the outcomes of the six-dimensional technique.

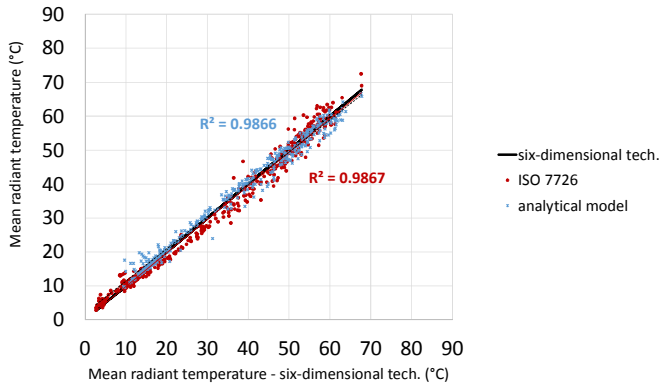

Fig. 10 Comparison among six-dimensional technique and both ISO7726 method and analytical model - average values assessed over a period of 10 minutes.

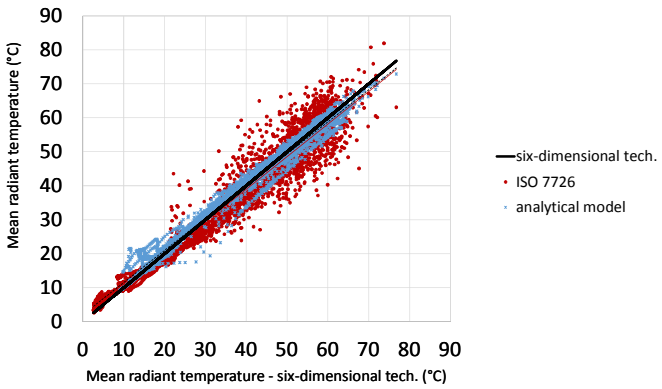

Fig. 11 Comparison among six-dimensional technique and both ISO7726 method and analytical model - average values assessed over a period of 1 minute.

On the other hand, when 1 minute average values are involved, the dispersion of the results yielded by the application of the ISO 7726 method enhances appreciably.

Supposedly, this phenomenon might be caused by the effect of the inertia of the globe-thermometer, which, within a period of 1 minute, couldn't be able to reach the required thermal equilibrium, especially when the solar radiation is involved.

On the contrary, the effects of this drawback are relatively constrained if 10 minute average experimental values are used. In this case, the effect of rapid changes in the radiation fluxes are inevitably smoothed and the probe seems be able to follow them rather consistently [29].

As far as the analytical model is concerned, its reliability in simulating the effect of solar radiation on MRT seems to be acceptable.

V. Conclusion

This paper aims to analyze the effective reliability of different methods used for the assessment of the mean radiant temperature, in outdoor urban environment.

Specifically, the globe-thermometer measurement method and the analytical model proposed in [23] and modified to make it suited to outdoor environment, were compared to the six-dimensional technique.

Results demonstrate that the globe-thermometer can be a suitable device for outdoor measurements of mean radiant temperature, provided that proper measure time steps are used.

As a matter of fact, when the average values of the involved parameters were gathered with a rate of 10 minutes, the globe method yielded results that were in good accordance to the outcome of the more accurate six-dimensional technique.

This is to be considered a positive outcome, considering that, albeit the six-dimensional technique is one of the most accurate measurement to obtain outdoor $\bar{T}_r$, it is rather expensive and does not allow multi-point surveys, because the needed equipment is quite immovable.

On the contrary, considering the time and space variability of the involved parameters, the use of a more streamlined procedure, like the one involving the globe, is desirable in order to make the process of MRT assessment more feasible, especially on large outdoor areas.

Supposedly, the reliability of the results might also be improved by furtherly broadening time step with which the parameters are collected. In this direction the development of the research is currently being planned.

Regarding the analytical model, its effectiveness in simulating the effect of solar radiation on MRT seemed to be reliable too. As a matter of fact, also in this case the results were rather consistent with the values estimated by means of the six-dimensional technique. However, further analysis, concerning more various environmental conditions in different seasons are needed for a definitive and more accurate judgment on this topic.

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